Mathematical Physics and Computation

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Curvenote

1 Linear Operator

1.1 Finite-dimensional Vector Space

1.1.1 Linear Operators

Change of basis Let V be a finite-dimensional vector space of dimesnion n. Let $\beta = \{e_1, ..., e_n\}$ and $\beta' = \{e'_1, ..., e'_n\}$ be both basis for V. We are interested in change reprentation of $x \in V$ from using β to β'. To do this we need to know how $e_j \in \beta$ can be represented using β':

$$e_\nu = a^\mu_\nu e'_\mu \tag{1}$$

Here coefficientcs a^μ_ν can be orgainized into a matrix. The matrix a^μ_ν looks like:

$$\begin{pmatrix} a^1_1 & a^1_2 & \cdots & a^1_n \\ \vdots & \vdots & & \vdots \\ a^1_n & a^1_n & \cdots & a^1_n \end{pmatrix} \tag{2}$$

Contravariant Transformation There are two interpretations of a^μ_ν:

1. **(a^μ_ν transform β' to β)** This is the definition above:

$$e_\nu = a^\mu_\nu e'_\mu \tag{3}$$

2. **(a^μ_ν transform representation in β to β')** Now for arbitrary $x \in V$, we have:

$$x = c'_\nu e'_\nu = c_\nu e_\nu = c_\nu(a^\mu_\nu e'_\mu) = (c_\nu a^\mu_\nu)e'_\mu \tag{4}$$

from which we recognize:

$$c'_\nu = (c_\nu a^\mu_\nu) \tag{5}$$

We note that the matrix has *opposite* effect on basis and representation. This is actually intuitive: in order to represent the vector in new basis β', we will have to write the old basis $e_\nu \in \beta$ using linear combinations of the new basis e'_μ. The later which effectively transforms basis in β' to β. For this reason, the representation is said to be transformed **contravariantly**.

Notation
The matrix a^μ_ν is commonly denoted by $[I]^{\beta'}_\beta$ in math textbooks. Therefore:

- Transform in represenation is denoted by:

$$[x]_{\beta'} = [I]^{\beta'}_\beta [x]_\beta \tag{6}$$

- Transform in matrix representation of operator is given by:

$$[T]_\beta = [I]^\beta_{\beta'} [T]_{\beta'} [I]^{\beta'}_\beta \tag{7}$$

Triangularization

Proposition 1.1 (Existence of eigenvalue in vector spaces on algebraically closed field)**.** *Suppose V is a finite-dimensional vector space over an algebraically closed field F, and $T \in \mathcal{L}(V)$, then T has at least one eigenvalue*

Theorem 1.1 (Algebraically closed field guarantees triangularizability)**.** *Suppose V is a finite-dimensional vector space over an algebraically closed field F, and $T \in \mathcal{L}(V)$, then T is always triangularizable.*

1.1.2 Inner Product Space

Note:The development of concepts need to keep extension to infinite analog in mind

The identity operator and the Kroncker's Delta Let V be a vector space endorsed with an inner product $\langle\cdot|\cdot\rangle$ of dimension n and $\beta = \{e_\nu\}$ be an orthonormal basis. Then we can compute the inner product between arbitrary $x, y \in V$ as follows:

$$\langle x|y\rangle = \langle e_\nu|x\rangle^* \langle e_\nu|y\rangle = \langle x|e_\nu\rangle\langle e_\nu|y\rangle = \langle x||e_\nu\rangle\langle e_\nu||y\rangle \tag{8}$$

But $\langle x|y\rangle = \langle x|I|y\rangle$, hence we get that

Lemma 1.2 (Outer-product decomposition of the identity operator)**.** *Let $\{e_\nu\}$ be an orthonormal basis for V, then:*

$$|e_\nu\rangle\langle e_\nu| = I \tag{9}$$

***Help!* What is happening here?**

Step 1 We first rewrite inner product as a sum of component-component product. In C^3 for instance,

$$\langle x|y\rangle = \sum_\nu (x^\nu)^* y^\nu = \sum_\nu \langle e_\nu|x\rangle^* \langle e_\nu|y\rangle \tag{10}$$

Note in the last step we envicted orthonormality of $\{e_\nu\}$. Adopting Einstein notaiton we get the first equality.

Step 2 We are bascially writing two consecutive inner product $\langle x, \alpha\rangle\langle\beta, y\rangle$ as a bilinear form $\langle x, Ay\rangle$ where $A = |\alpha\rangle\langle\beta|$. Consider this example in R^3:

$$\langle x, \alpha\rangle\langle\beta, y\rangle = \begin{pmatrix} x^1 & x^2 & x^3 \end{pmatrix} \begin{pmatrix} \alpha^1 \\ \alpha^2 \\ \alpha^3 \end{pmatrix} \begin{pmatrix} \beta^1 & \beta^2 & \beta^3 \end{pmatrix} \begin{pmatrix} y^1 \\ y^2 \\ y^3 \end{pmatrix} = \begin{pmatrix} x^1 & x^2 & x^3 \end{pmatrix} \begin{pmatrix} \alpha^1\beta^1 & \cdots & \alpha^1\beta^3 \\ \vdots & & \vdots \\ \alpha^3\beta^1 & \cdots & \alpha^3\beta^3 \end{pmatrix} \begin{pmatrix} y^1 \\ y^2 \\ y^3 \end{pmatrix} \tag{11}$$

In otherword, we are just invoking associative law(which holds in vector space):

$$(x\alpha^T)(\beta y^T) = x(\alpha^T\beta)y^T \tag{12}$$

where the outer product $\alpha^T\beta = A$.

We denote bilinear forms $x^T Ay$ in sandwich notations: $\langle x|A|y\rangle$

(Step 3) The only possble opeartor to be sandwitched without changing the result:

$$\langle x|Ay\rangle = \langle x|y\rangle \tag{13}$$

is when $A = I$. We therefore conclude that the outer product $|e_\nu\rangle\langle e_\nu|$ is the identity. In math notation, this is saying:

- For orthonormal basis $\{e_\nu\}$, we have

$$\sum_{\nu=1}^{n} e_\nu e_\nu^T = I \tag{14}$$

Now we consider representing $|x\rangle$ in β:

$$|x\rangle = (|e_\nu\rangle\langle e_\nu|)|x\rangle = |e_\nu\rangle\langle e_\nu|x\rangle \tag{15}$$

Now consider taking the inner product with another basis e_μ:

$$\langle e_\mu|x\rangle = \langle e_\mu|e_\nu\rangle\langle e_\nu|x\rangle \tag{16}$$

1.2 Infinite-dimensional Vector Space

1.2.1 Linear Operators

L^p Space

Definition 1.1 (space)**.** *L^p*
Let (X, Σ, μ) be a measure space and $1 \leq p \leq \infty$. The space $L^p(X, \mu)$ is the set of all measurable functions from X to R or C such that $\int_X |f(x)|^p d\mu < \infty$.

- *Note L^p is a normed space with norm:*

$$\|f\|_{L_p} = \left(\int_X |f|^p d\mu \right)^{1/p} \tag{17}$$

p can be the symbol ∞ with the norm defined as:

$$\|f\|_{L_\infty} = \inf\{C \in R_{\geq 0} : |f(x)| < C, \forall x\} \tag{18}$$

- *It is complete, hence it is a Banach Space*

Linear Operators

Definition 1.2 (Linear Operators)**.** *Let T be a map betwen two normed space X and Y. T is linear if:*

$$T(x + \lambda y) = T(x) + \lambda T(y) \tag{19}$$

$\forall x, y \in X, \lambda \in K$

If we consider linear functions $f : R \to R$, it seems continuity is free since we can always write $T(x) = kx$ hence:

$$\|T(x) - T(x_0)\| = \|k(x - x_0)\| = k\|x - x_0\| \tag{20}$$

so we can easily find a delta ball of radius $\frac{\epsilon}{k}$ given an epsilon ball such that the delta ball is mapped into the epsilon ball. More generally we can define bounded operator to make the above logic work:

Definition 1.3 (Bounded Operators)**.** *Let T be a map betwen two normed space X and Y. T is bounded if $\exists C > 0$:*

$$\|T(x)\|_Y \leq C\|x\|_X \tag{21}$$

$\forall x \in X$

Proposition 1.2 (Continuous and Bounded are equivalent for Linear Operator)**.** *The following are equivalent:*

1. *If X and Y are normed spaces and $T : X \to Y$ is a linear map which is continuous from X to Y*

2. *T is bounded*

Proof.

- Suppose T is **linear and bounded**, $\forall x_0 \in X$, $\forall \epsilon > 0$, consider the open ball around x_0 with radius ϵ B^o_ϵ. Then $\forall x \in B^o_\epsilon$:

$$\|T(x) - T(x_0)\| = \|T(x - x_0)\| \leq C\|x - x_0\| \tag{22}$$

Now to make $C\|x - x_0\| < \epsilon$, we need $\|x - x_0\| < \frac{\epsilon}{C}$. Hence $\delta(\epsilon) = \frac{\epsilon}{C}$

- Suppose T is **linear and continous**. Then in particular T is continous at $x = 0$. We can find a $\delta > 0$ such that $\|x\| < \delta \Rightarrow \|T(x)\| < 1$. Now to control arbitrary $x \in X$, we can rewrite x as

$$x = \left(\frac{2}{\delta}\right) \|x\| \left(\frac{\delta x}{2\|x\|}\right) = C\|x\|x' \tag{23}$$

Note $\|x'\| = \frac{\delta}{2}\|x\| < \delta$ hence $\|T(x')\| \leq 1$. By linearity $T(x) = T(C\|x\|x') = C\|x\|T(x')$. Thus:

$$\|T(x)\|_Y = C\|x\|\|T(x')\| \leq C\|x\| \tag{24}$$

□

It turns out all finite-dimensional linear operator are bounded. However, infinite-dimensional linear operator can easily be unbounded.

1.2.2 Linear Differential Operator

The Formal Linear Differential Operator We define the differentian opeartor as:

Definition 1.4 (Differentiation Opeartor). *Let $D : L^2[a,b] \to L^2[a,b]$ be the mapping defined by:*

$$D(u) = \frac{d}{dx}u \tag{25}$$

for all $u \in L^2[a,b]$

Now we can define the *formal linear operator* by some opeartor algebra:

Definition 1.5 (Formal linear differential operator). *Let $p(x)$ be any polynomial over C of positive degree:*

$$p(x) = a_\nu x^\nu \tag{26}$$

*where a_ν are functions of x. Define $L := p(D)$ as the **formal linear differential operator** that correspond to the **auxillary polynomial** p:*

$$L = a_\nu D^\nu \tag{27}$$

*The **order** of L is given by the order of p.*

We call it *formal* since we want to contrast with *concrete* linear differential opeartor which is a formal opeartor endorsed with boundary conditions(This convention is used in *Mathematical Physics by Michael Stone and Paul Goldbart*). We can think of the difference between them as:

- The **nullspace of formal opeartor** is the space of general solution of homogenous diffferential equation
- The **nullspace of concrete operator** is the space of particular solution to homogenous differential equation with boundary conditions

We'll explore the nullspace of formal operators with a bit more detail now.

Nullspace of formal operator We first observe something very nice about solutions to homogenous differential equations: they belong to C^∞(infinitely differentiable).

The Bootstrapping Lemma

Lemma 1.3. *Let L be a formal operator of order n. Suppose the coefficients of the auxillary polynomial are k times continously differentiable($a_\nu \in C^k$) then $\forall y \in \mathrm{Null}(L)$, y is at least $k+n$ times contiously differentiable.*

Proof. Suppose $y \in \text{Null}(L)$, note trivially $y \in C^n$. Then:

$$a_0 y + a_1 y^{(1)} + a_2 y^{(2)} ... + a_n y^{(n)} = 0 y^{(n)} = -(b_0 y + b_1 y^{(1)} + b_2 y^{(2)} ... \underbrace{b_{n-1} y^{n-1}}_{\in C^{\min(k,1)}}) \tag{28}$$

Where $b_\nu = a_\nu / a_n$. Now $b_\nu \in C^k$ by closedness under polynomial division. Also $y^{(\nu)} \in C^{n-\nu}$ hence the R.H.S is at least one time continously differentiable(the bottle neck being $a_{n-1} y^{(n-1)} \in C^{\min(k,1)}$). Therefore $y^{(n)} \in C^1$. But this implies $y \in C^{n+1}$ hence we repeat the above argument to get $y^{(\nu)} \in C^{n+1-\nu}$. The R.H.S has the bottleneck $a_{n-1} y^{(n-1)} \in C^{\min(k,n+1-(n-1))} = C^{\min(k,2)}$. We can repeat this procedure m times and yield that $y^{(n)} \in C^{\min(k,m)}$. This means the best we can get is $y^{(n)} \in C^k$ or $y \in C^{n+k}$(after which the coeficient becomes the bottleneck).

□

Corollary 1.3.1. *If the coefficients are in C^∞ then $y \in \text{Null}(L)$ are in C^∞.*

Nullspace of L is an n-dimensional subspace of C^∞

Linear Independence: Wronskian

Solving $Ly = 0$ with constant coefficients

$Ly = 0$ with general $a_\nu(x)$

1.2.3 Inner Product Space

Distribution and Test Functions

Continous Indexing

Duality Pairing

Exercises

Formal adjoint

Concrete Adjoint

The Sturm-Liouville Operator The **Sturm-Liouville operator** is a linear differential operator of the form

$$\mathcal{L} = -\frac{d}{dx}\left(p\frac{d}{dx}y\right) + q\frac{d}{dx} \tag{29}$$

We've shown that the Sturm-Liouville opeartor is self-adjoint with appropriate boundary condition in the sense that:

$$\langle u, \mathcal{L}v\rangle = \langle \mathcal{L}^* u, v\rangle \tag{30}$$

hence it is analgous to self-adjoint operator in finite-dimensional vector spaces, which as symmetric matrix representation when the underlying field is real. We now make this connection explicit, i.e., actually constructing the discretized, finite-dimensional matrix representation of the Sturm-Liouville Operator.

A Discrete Analog Let's consider a concrete SL opeartor defined on $[0, 1]$ with Dirchlet boundary condition $y(0) = y(1) = 0$. Instead of directly discretizing $\mathcal{L}$, we aim to find a matrix L such that discretizes the quadratic form:

$$F(y) = \int y\mathcal{L}y dx \to \hat{y}^T L\hat{y} \tag{31}$$

First we note that:

$$F(y) = \int_0^1 y\left(-\frac{d}{dx}\left(p\frac{d}{dx}y\right) + qy\right)dx = \int_0^1 -y\frac{d}{dx}\left(p\frac{d}{dx}y\right)dx + \int_0^1 qy^2 dx = -\left\{\left[yp\frac{dy}{dx}\right]_0^1 - \int p\left(\frac{dy}{dx}\right)^2\right\} + \int \tag{32}$$

Now we discretize y, p, q to N-dimensional arrays:

$$\hat{y} = [y_1, ..., y_N]\hat{p} = [p_1, ..., p_N]\hat{q} = [q_1, ..., q_N] \tag{33}$$

Using forward difference we get:

$$\hat{y} = \left(\frac{y_{i+1} - y_i}{h}\right)^2 \tag{34}$$

Rewrite the integral as a finite sum we obtain a discretized version of (32):

$$\boxed{\hat{F}(\hat{y}) = \sum_{i=1}^{N} hp_i \left(\frac{y_{i+1} - y_i}{h}\right)^2 + q_i y_i^2} \tag{35}$$

Our goal is now to find a matrix L such that $\hat{y}^T L \hat{y} = \hat{F}(\hat{y})$. Rewrite $\hat{F}(\hat{y})$:

$$\hat{F}(\hat{y}) = \sum_{i=1}^{N} Np_i(y_{i+1}^2 - 2y_{i+1}y_i + y_i^2) + \frac{1}{N} q_i y_i^2 \tag{36}$$

We can split $L = L_1 + L_2$ such that

1. $\hat{y}^T L_1 \hat{y} = \sum_{i=1}^{N} Np_i(y_{i+1}^2 - 2y_{i+1}y_i + y_i^2)$
2. $\hat{y}^T L_2 \hat{y} = \sum_{i=1}^{N} \frac{1}{N} q_i y_i^2$.

(Constructing L_1) Let's first deal with L_1. Consider the 2×2block with the upperleft corner at (i, i) that should represent $Np_i(y_{i+1}^2 - 2y_{i+1}y_i + y_i^2)$. This is easy:

$$A_i = Np_i \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \tag{37}$$

To assemble L_1 we can overlap A_i along the diagonal of L with a stride of 1. Concretely:

```
for i=1,...,N -1
    L[i,i] += Np_i
    L[i+1,i+1] += Np_i
    L[i,i+1] -= Np_i
    L[i+1,i] -= Np_i
```

Example 1.1. *A quick example for $N = 4$:*

$$4 \begin{pmatrix} p_1 & -p_1 & 0 & 0 \\ -p_1 & p_1 + p_2 & -p_2 & 0 \\ 0 & -p_2 & p_2 + p_3 & -p_3 \\ 0 & 0 & -p_3 & p_3 \end{pmatrix} \tag{38}$$

(Constructing L_2) L_2 is purely diagonal and is easier to derive. Since it only has quadratic term y_i^2, at each row i we only need to include y_i. Hence:

$$L_2 = \frac{1}{N} \begin{pmatrix} q_1 & & \\ & \ddots & \\ & & q_N \end{pmatrix} \tag{39}$$

(Assembly $L_1 + L_2$) It is clear that L should look like:

$$L = N \begin{pmatrix} p_1 & -p_1 & & & \\ -p_1 & p_1 + p_2 & -p_2 & & \\ & \ddots & \ddots & \ddots & \\ & & -p_{N-1} & p_{N-1} + p_N & -p_N \\ & & & -p_N & p_N \end{pmatrix} + \frac{1}{N} \begin{pmatrix} q_1 & & \\ & \ddots & \\ & & q_N \end{pmatrix} \tag{40}$$

However, since we are dealing with concrete operators with accompanying domain and boundary condition, we need to encoorperate the Dirchelt boundary condition $y(0) = 1, y(1) = 0$ into L. To do so we need to zero-out the first row and column of L. Hence finally we get:

Proposition 1.3. *The Sturm-Liouville Operator on* $[0, 1]$ *with Dirchlet boundary condition* $y(0) = y(1)$ *can be discretized as*

$$L = N \begin{pmatrix} 0 & 0 & & & & \\ 0 & p_1 + p_2 & -p_2 & & & \\ & -p_2 & \ddots & \ddots & -p_{N-1} & \\ & & \ddots & -p_{N-1} & p_{N-1} + p_N & 0 \\ & & & & 0 & 0 \end{pmatrix} + \frac{1}{N} \begin{pmatrix} 0 & & & & \\ & q_2 & & & \\ & & \ddots & & \\ & & & q_{N-1} & \\ & & & & 0 \end{pmatrix} \tag{41}$$

We can see that L is indeed a symmetric real matrix(Hermitian)!

Numpy Implementation This can easily be translated to naive `numpy` code:

```
class SturmLiouville:
    def __init__(self,N, p, q=None, grid='forward'):
        A = np.zeros(shape=(N,N))
        if grid == 'forward':
            for i in range(N -1):
                A[i, i] += p[i]
                A[i+1, i+1] += p[i]
                A[i, i+1] -= p[i]
                A[i+1, i] -= p[i]

        if q is not None:
            I = np.eye(N)
            L = A + q @ I
        else:
            L = A

        # Dirchlet Boundary Condition
            L[0,0]=0.0
```

```
            L[0,1]=0.0
            L[1,0]=0.0
            L[N -1,N -1]=0.0
            L[N -1,N -2]=0.0
            L[N -2,N -1]=0.0
        self.L = (L * N ** 2)
        self.N = N

    def solve_eigen(self, top_k=10):
        eval, evec = np.linalg.eig(self.L)
        top_k_idx = np.argsort(eval[eval>0])[:top_k]
        return eval[top_k_idx] , evec[:, top_k_idx], top_k_idx
```

Example: $-\frac{d^2}{dx^2}$ as a Matrix We can do a quick example to showcase that L is indeed a discretization of $\mathcal{L}$.

Example 1.2 (-). $\frac{d^2}{dx^2}$
We have already seen that the opeartor:

$$T = -\frac{d^2}{dx^2}, D(T) = \{y, Ty \in L^2[0,1] : y(0) = y(1) = 0\} \tag{42}$$

has eigenvalues and eigenvectors:

$$y_j(x) = \sqrt{2}\sin j\pi x \lambda_j = j^2\pi^2 \tag{43}$$

for $j = 1, 2, 3....$ The problem is that, can we solve a matrix eigenvalue problem to recover exactly the same result? Indeed we can! Passing $p = -1$ and $q = 0$ gets us:

$$L = N\begin{pmatrix} 0 & 0 & & & & \\ 0 & 2 & -1 & & & \\ & -1 & \ddots & \ddots & & \\ & & \ddots & 2 & -1 & \\ & & & -1 & 2 & 0 \\ & & & & 0 & 0 \end{pmatrix} \tag{44}$$

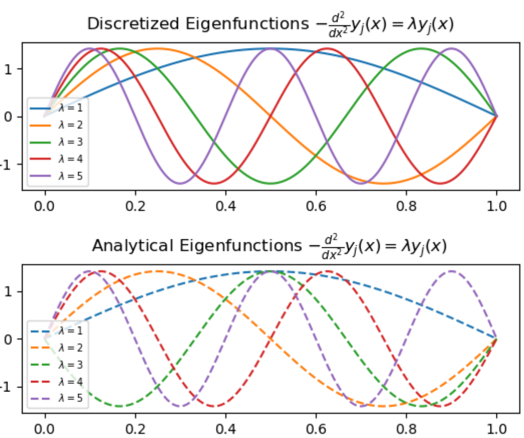
and solving the matrix eigenvalue problem

$$L\hat{y} = \lambda\hat{y} \tag{45}$$

gets us: [1]

[1]Since iterative eigensolvers gives eigenvectors of arbitrary magnitude $v_j = a\sin(j\pi(x))$, we need to normalize properly by 1.Ensure $a > 0$

1. Divide by $\sqrt{\sum y_i^2 h}$


Discretized Eigenfunctions $-\frac{d^2}{dx^2}y_j(x) = \lambda y_j(x)$
$\lambda = 1$
$\lambda = 2$
$\lambda = 3$
$\lambda = 4$
$\lambda = 5$
Analytical Eigenfunctions $-\frac{d^2}{dx^2}y_j(x) = \lambda y_j(x)$
$\lambda = 1$
$\lambda = 2$
$\lambda = 3$
$\lambda = 4$
$\lambda = 5$

2 Calculus of Variation

2.1 Euler-Lagrange Equation

2.1.1 Differential Equation from Physics

A lot of physical science and engineering is about finding a reduction from a physical problem to differential equations. The newtownian way is to zoom into some local point of the system and derive a set of constraint that holds in the neighborhood. In the case of the vibration of a string, for example, we would find some local segment of string and analyze the tension that acts on it:

$$T(\sin(\theta_1) - \sin(\theta_2)) = m\frac{\partial^2 u}{\partial t^2} \tag{46}$$

where u is the vertical displacement from equilibrium. We then take the calculus limit and find out that locally the accelearation of the vertical displacement should be proportional to the curvature. Hence the one-dimensional wave equation:

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} \tag{47}$$

[2]

The Lagrangian way is the opposite, we recognize some global objective the system need to be optimized toward, i.e., minimizing the lagragian $L = T - V$ where T is *kinetic energy* and V is *potential energy.* Then, by introduce some perturbation, we extract a constraint which also manifest itself as a differential equation. The following sections descirbes this process in detail.

Newtonian and Lagrangian Mechanics

- Newtonian mechanics compute the balance of forces locally and obtain differential equation through

$$F = m\frac{d^2\mathbf{x}}{dt^2} \tag{48}$$

- Lagrangian mechanics translate the physics into some global optimization problem and reduce to differential equation through the Euler-Lagrange equation

$$\frac{\partial L}{\partial y} - \frac{d}{dx}\frac{\partial L}{\partial y'} = 0 \tag{49}$$

[2]Here a is a physical parameter intrinsic to the string, i.e., $\sqrt{T}\rho$ where T is the tension at equilibrium adn ρ is the density.

2.1.2 Functional : A global optimization objective

A functional in *calculus of variation* is a map from a function to a real number, i.e., $J : C^\infty \to R$. In particular, it mostly comes in some sort of a integral. The simplest form being analgous to that of the lagrangian:

$$J[y] = \int_{x_1}^{x_2} f(x, y, y')dx \tag{50}$$

The above example suffice to establish the explict connection of (50)to Lagrangian:

Example 2.1 (The lagrangian for the vibrating string)**.** *Let u be deviation from the equilibrium position*

1. ***The kinetic energy:***

$$T = \frac{1}{2}\int_0^L \rho u_t(x,t)^2 dx \tag{51}$$

where L is the length of the string

2. ***The potential energy:***

$$V = \frac{1}{2}k\int_0^L 1 + u_x^2 dx \tag{52}$$

[3]

The lagrangian is hence:

$$L = \int_0^L \frac{1}{2}\left(\rho u_t(x,t)^2 - k(1+u_x^2)\right) \tag{55}$$

We now scrutinize how exactly we optimize (50).

---

[3] It is tempting for some of us who are not trained as a physics student to write the potential energy as $\frac{1}{2}\left(\int_a^b \sqrt{1+u_x^2}dx\right)^2$. The rationale is as follows: if we pause at a moment and straigten the vibrating string as a line. Then we observe how much elongation had happened and use hooke's law on the entire string. This is physically wrong since the elongation happens inhomogenously hence we have to apply hooke's law locally and integrate. Also mathematically,

$$\left(\int f\right)^2 = \int f^2 \tag{53}$$

iff f is some constant function. The general relatinoship can be obtained from Cauchy-Shwarz inequality by taking $f = \sqrt{1+u_x^2}$ and $g = 1$

$$\left(\int_0^L \sqrt{1+u_x^2}\right)^2 \leq \int_0^L 1 + u_x^2 dx \tag{54}$$

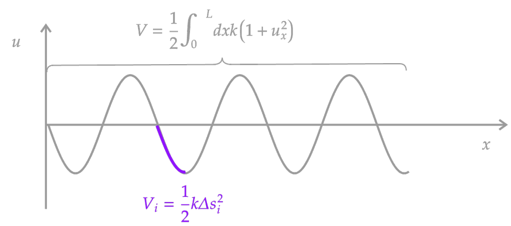


Figure 1: *
The potential energy of a vibrating string.

2.1.3 Euler-Lagrange Equation

Theorem 2.1 (Euler-Lagrange Equation)**.** *y is a stationary point for*

$$J[y] = \int_{x_1}^{x_2} f(x, y, y^{(1)}, ..., y^{(n)})dx \tag{56}$$

iff:

$$\boxed{\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0} \tag{57}$$

The boxed equation is known as the ***Euler-Lagrange Equation***

2.1.4 Discretized Euler-Lagrange Equation

2.1.5 First Integral

Definition 2.1 (First Integral)**.** *Suppose*

$$y^{(n)} = f(x, y, y^{(1)}, ..., y^{(n-1)}) \tag{58}$$

is an ordinary differential equation of order n. We can define an equivalent n-dimensional system by letting:

$$y_1 = yy_2 = y^{(1)} \vdots y_n = y^{(n-1)} \tag{59}$$

Denote by:

$$\frac{dy_i}{dx} = f_i(x, y_1, ..., y_n) \tag{60}$$

Now suppose f is:

1. *Continous about $x, y_1, ..., y_n$*

2. Continuously differentialbe about $y_1, ..., y_n$

Define the first integral of the n^{th}-order problem to be a function $I = I(x, y_1, ..., y_n)$ that also satisfies the above two requirements , and furthermore has the definitive property:

$$I(x, y_1, ..., y_n) = Constant \tag{61}$$

along solution curves $\Gamma = (y_1(x), y_2(x), ..., y_n(x))$ with the constant determined by the solution curve.

We are confronted by the higher-order ordinary differential equation:

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0 \tag{62}$$

The compute the first integral of this equation is to rewrite the above to obtain an equation of the form:

$$\frac{d}{dt} I(x, y, y', ...) = 0 \tag{63}$$

The strategy is to compute the difference between (62) and the exact differential:

$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \sum_{j \geq 0} \frac{\partial f}{\partial y^{(j)}} \frac{dy^{(j)}}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{dy}{dx} + \frac{\partial f}{\partial y^{(1)}} \frac{dy^{(1)}}{dx} + \cdots \tag{64}$$

If we multiply (62) by y' and subtract the two equations we get:

$$\frac{df}{dx} - y'\left\{\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right)\right\} = \frac{\partial f}{\partial x} + y''\frac{\partial f}{\partial y'} + y'\frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) - \sum_{j \geq 2} \frac{\partial f}{\partial y^{(j)}} \frac{dy^{(j)}}{dx} \tag{65}$$

The particular form that allows easy first integral

If we assume f is only dependent on y, y', the above becomes $f(y, y')$, then

$$\sum_{j \geq 2} \frac{\partial f}{\partial y^{(j)}} \frac{dy^{(j)}}{dx} = 0 \frac{\partial f}{\partial x} = 0 \tag{66}$$

Now:

$$\frac{df}{dx} - \underbrace{y'\left\{\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right)\right\}}_{0} = \frac{d}{dx}\left(y'\frac{\partial f}{\partial y'}\right) \tag{67}$$

Hence:

$$\frac{d}{dx}\left(f - y'\frac{\partial f}{\partial y'}\right) = 0 \tag{68}$$

We have sucesfully obtained the first inetgral:

$$I(y, y) = f - y'\frac{\partial f}{\partial y'} \tag{69}$$